

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech (EC) Examination****May 2016****EC 201/EC201.01 Digital Electronics & Logic Design****Date: 04.05.2016, Wednesday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q - 1 Answer the following questions. [07]**

- (a) Fill the blank: $(B65F)_{16} = (\quad)_{10}$ [02]
- (b) Fill the blank: $(41)_{10} = (\quad)_2$ [01]
- (c) Using 10's complement, subtract $72532 - 3250$. [02]
- (d) Which code is used in K-Map? [01]
- (e) Define: Combinational Digital Circuits. [01]

Q - 2 Answer the following questions. [14]

- (a) Express the Boolean Function $F = A+B'C$ in a sum of min terms and in product of max terms form. [07]
- (b) Draw & Explain two inputs (i) AND (ii) OR and (iii) EX-OR gates. [07]

OR**Q - 2 Answer the following questions. [14]**

- (a) Reduce using K-Map $\sum m(0, 2, 6, 10, 11, 12, 13) + d(4, 5, 14, 15)$. [07]
- (b) Implement $F(A, B, C, D) = \sum m(0, 1, 3, 4, 8, 9, 15)$ using multiplexer, choose A as input line. [07]

Q - 3 Answer the following questions. [14]

- (a) Implement Full Adder with 3x8 decoder. [07]
- (b) Design a 4 bit synchronous UP / DOWN counter. [07]

OR**Q - 3 Answer the following questions. [14]**

- (a) Explain carry propagation in detail. [07]
- (b) Explain Johnson Counter. [07]

SECTION – II

Q - 4 Answer the following questions. [07]

- (a) Simplify the following Boolean equation $F(A,B,C)=ABC+AB'+ABC'$ [02]
- (b) Prove: $AB+A'C+BC=AB+A'C$ [02]
- (c) Design AND, OR and NOT gate using any one of universal gate. [03]

Q - 5 Answer the following questions. [14]

- (a) Design 4-bit BCD to Excess-3 code conversion using basic logic gates. [07]
- (b) Convert S-R Flip flop to D Flip flop. [07]

OR

- (b) Write down excitation tables for S-R, J-K, D and T flip flops with logic diagram. [07]

Q - 6 Answer the following questions. [14]

- (a) Write a short note on 4-bit Universal shift register. [07]
- (b) Explain 4 bit Magnitude Comparator with logic diagram and functions. [07]

OR

- (b) Explain read and write functions related to memory unit. [07]